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Investigation on Sensing and Stability of Nitrogen-doped ITO Thin-film Strain Gauges at High Temperatures

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Abstract

Nitrogen-doped indium tin oxide (ITO) has been applied in the thin-film strain gauges *(TFSGs)* due to their high stability, excellent piezoresistivity and antioxidation at elevated temperatures. However, the mechanism on the sensing and stability of the nitrogen-doped ITO *TFSGs* at high temperatures was not comprehensively clarified. In this work, various ITO *TFSGs* were fabricated by *RF* magnetron sputtering with different nitrogen partial pressures (*NPPs*) of 5%~40%. The elemental composition and band structures of the ITO *TFSGs* were examined by the energy dispersive X-ray spectroscopy (EDS) and X-ray photoelectron spectroscopy (XPS), respectively. Results show that the Fermi energy level shifts closer to the valence band maximum energy (E_v) gradually with the growth of *NPPs*, causing a reduction in the number of electrons ionized to the conduction band. The smallest content change rates of nitrogen (3.8%) and oxygen (1.6%) after subjecting to the thermal strain test were observed in the 20%N_2 ITO *TFSG*. In consequence, the 20%N_2 ITO *TFSG* exhibits the lowest resistance drift rate (*DR*) at high temperatures due to its stable elemental composition. Moreover, we found that the band structures and elemental composition of the ITO *TFSGs* are the main factors affecting their piezoresistive response at different temperatures. The band structures play a major role in the gauge factors (*GFs*) of the ITO *TFSGs* at room temperature and 600 ℃. The element variation takes responsibility for the different *GFs* of the ITO *TFSGs* at 800℃, 900℃ and 1000℃. In addition, the piezoresistive stability is also dependent on the elemental composition affected by the dynamic equilibrium between the diffusion amount of oxygen and the escape number of the nitrogen in the ITO thin films at high temperatures.

1. Introduction

Ultra-high temperature thin-film sensors (*UTTSs*) have been widely used to measure the structural behavior of the hot sections in harsh environments including elevated temperature, high pressure and rigorous mechanical loading[1–3]. The *UTTSs* can be directly fabricated on the surface of the hot components since they are non-intrusive for the multiple physical parameters measurements such as strain[4–6], temperature[7–10] and heat flux[11–13]. As one of the *UTTSs*, thin-film strain gauges (*TFSGs*) are robust enough to in-situ measure the strain of hot components in harsh environment since they have negligible mass and considerably less thickness than that of the gas-phase boundary layer[14, 15]. Therefore, the surface airflow fields and vibration patterns will be minimally disturbed by these sensors. Recently, indium tin oxide (ITO) has been applied in the *TFSGs* due to its high temperature endurance, excellent piezoresistivity and antioxidation at elevated temperatures. However, ITO *TFSGs* exhibit resistance drift with the time at high temperatures, which will limit the improvement of the measurement accuracy and stability [1].

Literature shows that the piezoresistivity of the ITO *TFSGs* was conditional upon their microstructures and electrical properties, whereby these characteristics were significantly affected by their elemental composition. For example, Fang and his coworkers prepared the ITO *TFSGs* with different oxygen concentrations using pulsed laser deposition[16]. Their results suggest that the piezoresistivity of ITO

TFSGs increased with the growth of oxygen content. Tao You et al introduced aluminum into the ITO films by co-sputtering with Al_2O_3 and ITO targets to investigate the stability of the aluminum-doped ITO *TFSGs* at high temperatures[17]. Results indicate that aluminum-doped ITO *TFSGs* have good thermal stability at elevated temperatures up to 1450℃. However, it is hard to achieve homogeneous doping in the ITO *TFSGs*, which will affect their property uniformity.

Some researchers had reported that nitrogen-doped ITO *TFSGs* could have more stable piezoresistive performance than undoped ITO *TFSGs*. Gregory et al prepared ITO strain sensors using reactive sputtering in nitrogen-rich plasma ambient[18]. Compared with the ITO *TFSGs* prepared in an argon/oxygen environment, the nitrogen-doped sensors exhibited better performance in electrical stability and piezoresistive response at high temperatures due to the formation of the controlled nano-porosity. Yang and his co-authors fabricated the ITO *TFSGs* using magnetron sputtering under argon/oxygen ambient and introduced nitrogen into the ITO *TFSGs* via heat-treatment in nitrogen [19, 20]. Their results suggest that the nitrogen-doped ITO *TFSGs* had a smaller but more stable gauge factor (*GF*) than undoped sensors. However, these methods cannot control the nitrogen doping content and are difficult to accurately modify the performance of the ITO *TFSGs.* The effect of different nitrogen doping concentration on the piezoresistive stability and the *GFs* of the ITO *TFSGs* have not been comprehensively illuminated in those works. More details between the nitrogen incorporation and the piezoresistive response of ITO *TFSGs* still need further clarification.

The purpose of this paper is to further investigate the effect of nitrogen doping concentration on the sensing and stability, various ITO *TFSGs* were prepared using *RF* magnetron sputtering with different N_2/Ar ratios ranging from 5%~40%. The elemental composition and band structure of the ITO *TFSGs* were examined, and their piezoresistive response was also recorded. The influence mechanism of the nitrogen-doping content on the piezoresistive stability and the *GFs* of the ITO *TFSGs* were revealed. Results show that 20%*NPP* was the optimal nitrogen incorporation parameter due to its high stability and high *GF* at elevated temperatures

2. Exprimental Details

2.1 *TFSGs* preparation

Different ITO *TFSGs* were prepared on high-purity alumina substrates using MEMS fabrication technology, as shown in Fig.1. The alumina substrates were cut into the constant strain beams according to the designed shape and size (Fig.2d), which can provide adequate electrical insulation at elevated temperatures. The alumina constant strain beams were ultrasonic cleaned successively with Acetone, Ethanol, and Distilled Water and blow-dried by nitrogen (Fig.1a). They were then baked at 150℃ for 120 minutes prior to the *TFSGs* preparation. Subsequently, different ITO films were deposited onto the substrates using MW-ECR plasma enhanced magnetron sputtering system with a high-purity ITO target (99.99 wt.%, *Φ*68*3mm) comprised of 90 wt.% Indium oxide and 10 wt.% Tin oxide (Fig.1b). The sputtering parameters are listed in Table.1. The mixed gas with different *NPPs* of 5%~40% was leaked

into the sputtering chamber and working pressure was set to 0.6 Pa. The sputtering time is one hour. After the deposition, these sputtered films were spin-coated with a positive photoresist (BP212, Kempur Microelectronics Ins, Beijing, China) at a speed of 2400r/min (Fig.1c). The photoresist was soft baked at 85℃ for 30minutes (Fig.1d), followed by patterning with a film mask via contact exposure (Fig.1e) and developing in 0.5 wt.% sodium hydroxide solution for the 30s (Fig.1f). Subsequently, these patterned samples were etched in Aqua regia to delineate the final *TFSGs* structures (Fig.1g). Ultimately, the patterned *TFSGs* were placed in acetone to remove the residual photoresist (Fig.1h). The ITO *TFSGs* were heat-treated at 1000°C in a programmable muffle furnace (TongKe, TCXQ-1700, China) for two hours.

Table 1 Sputtering parameters of the ITO TFSGs under different NPPs.

Nitrogen partial pressure (%)	The flow ratio between N_2 and Ar (sccm)	Working power (W)	Base pressure (Pa)	Working pressure (Pa)	Sputtering temperature (℃)
5%N_2 ITO *TFSG*	1/19	300	3×10^{-3}	0.6	Room temperature
10%N_2 ITO *TFSG*	2/18				
20%N_2 ITO *TFSG*	4/16				
30%N_2 ITO *TFSG*	6/14				
40%N_2 ITO *TFSG*	8/12				

2.2 Characterization and static strain testing

The valence band structures of the various ITO *TFSGs* were examined by X-ray photoelectron spectroscopy (XPS, K-Alpha+, America) calibrated using C1s line (284.8 eV). The energy dispersive X-ray spectroscopy (EDS, NOVA Nano SEM450, FEI, America) was used to analyze the elemental composition before and after the thermal strain test. The piezoresistive response of the different ITO *TFSGs* was tested in a high-temperature furnace programmed from room temperature to 1000℃, as shown in Fig.2a-2c. The furnace was ramped at 30℃/min to the specified temperature and maintain for at least 30 minutes to establish thermal equilibrium. The strain was induced by an alumina rod driven by a linear motor to compress the constant strain beam fixed by the Al_2O_3 ceramic clamping. A laser displacement sensor was used to record the applied strain of the constant strain beams by measuring their deflection and interfaced to an acquisition card via an I/O board. The corresponding resistance variations of the ITO *TFSGs* were noted using a digit multimeter (DMM6500 6½, TEKTRONIX, America).

The piezoresistive response of the ITO *TFSGs* is described by gauge factor (*GF*) as shown in equation (1) , which is dependent on the initial resistance and the resistance change ($\triangle R$) induced by the applied strain (ε).

$$GF=\frac{\triangle R}{R_{ref}}\times\frac{1}{\varepsilon}=(1+2\mu)+\frac{d\rho/\rho}{\varepsilon} \quad (1)$$

In addition, the stability of the ITO *TFSGs* is evaluated by drift rate (*DR*), and it is defined as equation (2):

$$DR=\frac{\triangle R}{R_{ref}}\times\frac{1}{\triangle t} \quad (2)$$

Where R_{ref} is the initial resistance at a specified testing temperature, μ is the Poisson ratio related to the geometric dimension of *TFSGs*, ρ is the resistivity. Generally, μ is a constant. Therefore, the *GF* of ITO *TFSGs* mainly depends on the variation of resistivity, which is determined by the carrier concentration and mobility at elevated temperatures. $\triangle t$ is the time during the test of resistance change $\triangle R$.

3. Results And Discussion

3.1 Characterizations of the ITO *TFSGs*

Fig.3a shows the XPS valence band spectra of the different ITO *TFSGs*, and the mechanism of nitrogen incorporation is revealed. A peak at the binding energy of 4.65eV is observed, which might be due to the In5p-O2p interaction and correspond to the orbital state of p-like valence band[21, 22]. The peak at 4.65 eV shifts to 3.5 eV with the increase of *NPPs,* which can be attributed to a weaker electronegativity of nitrogen (3.04) than that of oxygen (3.44). This can be deduced that more nitrogen atoms were incorporated into the ITO *TFSGs* and substituted a portion of oxygen atoms with the increase of *NPPs*. The valence band maximum energy (E_v) of the ITO *TFGSs* was obtained from the linear fitting of the leading edge of the valence band and the flat energy distribution based on the XPS data, and finding the crossover point of these two lines[23]. As shown in Fig.3a, the E_v of the ITO *TFSGs* decreases from 2.88 to 1.85 eV when the *NPP* increases from 5% to 40%. Intrinsic ITO was produced by introducing tin into indium oxide, which forms a donor impurity level E_d, as shown in Fig3b. Each tin atom doping in In_2O_3 has a valence of four and indium is three, which can contribute one electron to the conduction band. When nitrogen atoms replace oxygen atoms in the process of nitrogen doping, each nitrogen atom will produce an electron hole. In consequence, an acceptor impurity level *Ea* was formed in the nitrogen-doped ITO *TFSGs*, as shown in Fig3c. The donor impurity level E_d will compensate a part of the acceptor impurity level E_a, which causes the Fermi energy level (E_F) of the nitrogen-doped ITO TFSGs to shift closer to the E_v with the increase of *NPPs*[22], as shown in Fig.3c. Moreover, according to the results in the literature, the nitrogen incorporation will not influence the band gap(E_g, as shown in Fig3b and 3c) of ITO[24]. In that case, the energy difference between the Fermi energy level (E_F) and the conduction band

(E_c) will become larger with the increase of *NPPs*. Therefore, this will cause a reduction of the carrier concentration due to the decrease of the number of electrons ionized to the conduction band in the ITO *TFSGs* with the growth of *NPPs*.

The elemental composition is an important factor affecting the piezoresistive performance. Fig.4 shows the nitrogen and oxygen content of the ITO *TFSGs* prepared with various *NPPs* before and after the thermal strain test. Herein, the element content is defined as the proportion of nitrogen and oxygen in the sum of the four elements containing nitrogen, oxygen, indium and tin. Initially, the nitrogen content decreases and the oxygen content of the ITO *TFSGs* grows with the increase of *NPPs* before the thermal strain test. After the thermal strain test from room temperature to 1000℃, the nitrogen content declines (Fig 4a) and the oxygen content increases at different degrees (Fig 4b). This is because some metastable state nitrogen will escape from the grain boundary and the originally occupied oxygen vacancies at high temperatures [18]. On the other hand, more oxygen in the air will also diffuse into the ITO *TFSGs* at elevated temperatures [20]. Consequently, the unbalance between these two processes will affect the content of nitrogen and oxygen in the ITO *TFSGs*. Notably, the smallest content change rates of nitrogen (3.8%) and oxygen (1.6%) among the ITO *TFSGs* are observed in the 20%N_2 ITO *TFSG*, suggesting this *TFSG* has the most stable chemical composition.

3.2 Strain testing

The piezoresistive responses of the ITO *TFSGs* at room temperature are shown in Fig.5a-e. The resistances of all the ITO *TFSGs* decrease with the growth of the applied strain and increase with the decline of the strain. All the ITO *TFSGs* exhibit excellent stability in the resistance response during three loading and unloading cycles, and almost no resistance drift (maximum drift rate is only 0.0059/h) is found. Herein, the *GFs* of the ITO *TFSGs* were calculated according to equation (1). The *GFs* of the ITO *TFSGs* decrease from 2.78 to 1.94 with the increase of the *NPPs*, as shown in Fig5f. This can be attributed to the larger energy difference between the Fermi energy level (E_F) and the conduction band (E_c) with the increase of *NPPs*, causing fewer ionized electrons transfer to the conduction band in the ITO *TFSGs* under the same strain.

The furnace was then ramped from room temperature to 600℃. The piezoresistive responses of the ITO *TFSGs* at 600℃ are shown in Fig.6a-e, and the resistances can negatively change in pace with the applied strain. Compared with the testing results at room temperature, the different resistance drift appears in the ITO *TFSGs* at 600℃. As shown in Fig.6f, the *DRs* of the ITO *TFSGs* will initially decrease from 0.027/h to 0.0015/h and then grow from 0.0015/h to 0.02/h with the increase of the *NPPs*. This is because both inadequate and excessive nitrogen doping concentration will influence a dynamic equilibrium between the nitrogen escape from the ITO *TFSGs* and the oxygen diffusion into the *TFSGs.* Therefore, this will cause a destabilization of piezoresistive response. Meanwhile, it also suggests that the piezoresistivity of the 20%N_2 ITO *TFSG* is the most stable one at 600℃. In addition, different ITO *TFSGs* have various *GFs*, as shown in Fig.6f. The *GF* declines from 2.79 to 1.48 with the enhancement of *NPPs* due to the decrease of carrier concentration, which depends on the decline of the amounts of

electrons ionized to the conduction band since the shift of E_F in the ITO *TFSGs.* However, the *GF* of 10%N_2 ITO *TFSG* is smaller than that of the 20%N_2 ITO *TFSG.* This is because the E_F of these two *TFSGs* is similar. In that case, the Hall mobility might decrease when the temperature was raised to 600℃ due to the enhanced scattering effect induced by the unbalance between the nitrogen escape and the oxygen diffusion occurred in the 10% N_2 ITO TFSG[25]. Thus, this will play a dominant role in the smaller *GF* of the 10% N_2 ITO *TFSG* than that of the 20% N_2 ITO *TFSG.*

The furnace was continually ramped from 600℃ to 800℃ and the piezoresistive responses of the ITO *TFSGs* at 800℃ are shown in Fig.7a-e. The resistances of all the ITO *TFSGs* changed synchronously with the applied strain, and the variation trend of resistance is consistent with the results discussed in Fig5 and 6. However, the piezoresistive stability decreases with the increase of temperature and the high resistance drift also appears in the ITO *TFSGs*. As shown in Fig.7f, all the *DRs* of the ITO *TFSGs* at 800℃ are larger than those at 600℃. Notably, the *DR* of the 20%N_2 ITO *TFSG* is the lowest at 0.015/h compared with other *TFSGs* at 800℃, suggesting the *TFSG* prepared with 20% *NPP* still presents the most stable piezoresistivity. This further indicates that the nitrogen doping concentration should be moderate. Slight or redundant nitrogen content in the ITO *TFSGs* is not conducive to the stability of piezoresistive response at high temperatures. In addition, the ITO *TFSGs* also have a great difference in the *GFs,* as shown in Fig.7f. The *GFs* of the ITO *TFSGs* no longer monotonically changes with the increase of *NPPs* at 800 ℃. The *GF* of the 20%N_2 ITO *TFSG* is the highest at 2.66 among all the *TFSGs*. Comparatively, the *GFs* of the 5%N_2 and 10%N_2 ITO *TFSGs* are lower than those of 30%N_2 and 40%N_2 ITO *TFSGs.* This might be attributed to the disequilibrium between the amount of nitrogen escape and the number of oxygen diffusion in the ITO *TFSGs*. The oxygen diffusion amount of the 5%N_2 and 10%N_2 ITO *TFSGs* is greater than their amount of nitrogen escape, while the unbalanced relationship will inversely exist in the 30% N_2 and 40% N_2 ITO *TFSGs.* Thus, this will result in the lower carrier concentration and the smaller resistance change in the 5%N_2 and 10%N_2 ITO *TFSGs* than those in the 30% N_2 and 40% N_2 ITO *TFSGs*. In conclusion, the carrier concentrations affected by the element variation become dominant in the piezoresistive response of the different ITO *TFSGs* and the band structures is no longer take effect at 800℃.

The strain test results of the ITO *TFSGs* at 900℃ are shown in Fig.8a-e. It can be seen that the piezoresistive stability of the ITO *TFSGs* at 900℃ are similar to that of the ITO *TFSGs* at 800℃. The *DR* of the 20%N_2 ITO *TFSG is* also the lowest one (0.013/h), which is consistent with the results at 800℃. In addition, the *GFs* of the ITO *TFSGs* also bear resemblance to the results at 800℃. The 20% ITO *TFSG* has the highest *GF* of 2.42 among all the *TFSGs* at 900℃, as shown in Fig.8f. The reasons for the change trends of the *DR*s and *GFs* in the ITO *TFSGs* are the same as those have been discussed at 800℃, which suggests that the ITO *TFSGs* have the similar piezoresistive performance at 800℃ and 900℃.

The testing temperature was ultimately raised to 1000℃, and the resistance changes of the ITO *TFSGs* with the applied strain are shown in Fig.9a-e. The waveform of the piezoresistive response has severely degraded. The *DRs* of the ITO *TFSGs* at 1000℃ increase significantly compared with those of the *TFSGs*

below 900℃ (Fig.9f). This is because the electrical properties were seriously influenced by the elemental composition at 1000℃. More oxygen in the air was activated to diffuse into the ITO *TFSGs* at 1000 ℃ and occupied oxygen vacancies, leading to the compensation of the doubly charged oxygen vacancies and the decline of the carrier concentration [20]. At the same time, some metastable nitrogen incorporated in the ITO *TFSGs* will diffuse out and some eventually trapped at triple junctions and grain boundaries [18]. In consequence, these two processes will result in the decline of nitrogen content (Fig 4a) and the increase of oxygen content (Fig 4b), exacerbating the chaos of the carrier concentration and Hall mobility of the *TFSGs* at 1000℃. This will lead to the severe destabilization of the piezoresistive response. However, the 20%N_2 ITO *TFSG* can still maintain the relatively low *DR* of 0.015/h due to the smallest content change rates of nitrogen (3.8%) and oxygen (1.6%), which indicates the superiority of this *TFSG* in piezoresistive stability at high temperatures. Moreover, the *GF* of the 20%N_2 ITO *TFSG* is also the highest one compared with other *TFSGs* because of its stable elemental composition.

4. Conclusion

This paper aims to investigate the effect of the *NPPs* on the piezoresistive response of the ITO *TFSGs* from room temperature to 1000℃. Different ITO *TFSGs* were prepared by *RF* magnetron sputtering with various N_2/Ar ratios of 5%~40%. The XPS valence band spectra of the ITO *TFSGs* suggest that the E_F shifts to the E_v with the growth of *NPPs* due to the compensation between the donor impurity level (E_d) and acceptor impurity level (E_a). The elemental analysis indicates that the 20%N_2 ITO *TFSG* has the smallest content change rates of nitrogen (3.8%) and oxygen (1.6%) after the thermal strain test. Consequently, the 20%N_2 ITO *TFSG* exhibits the lowest *DR* at high temperatures due to its stable elemental composition. In addition, the mechanism of the piezoresistive response in the different ITO *TFSGs* at high temperatures was revealed. Results show that the band structures play the main role in the *GFs* of the ITO *TFSGs* at room temperature and 600 ℃. The element variation is responsible for the different *GFs* of the ITO *TFSGs* at 800℃, 900℃ and 1000℃. The piezoresistive stability also depends on the elemental composition influenced by the dynamic equilibrium between the diffusion amount of oxygen and the escape amount of nitrogen in the ITO *TFSGs* at high temperatures.

Declarations

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Figures

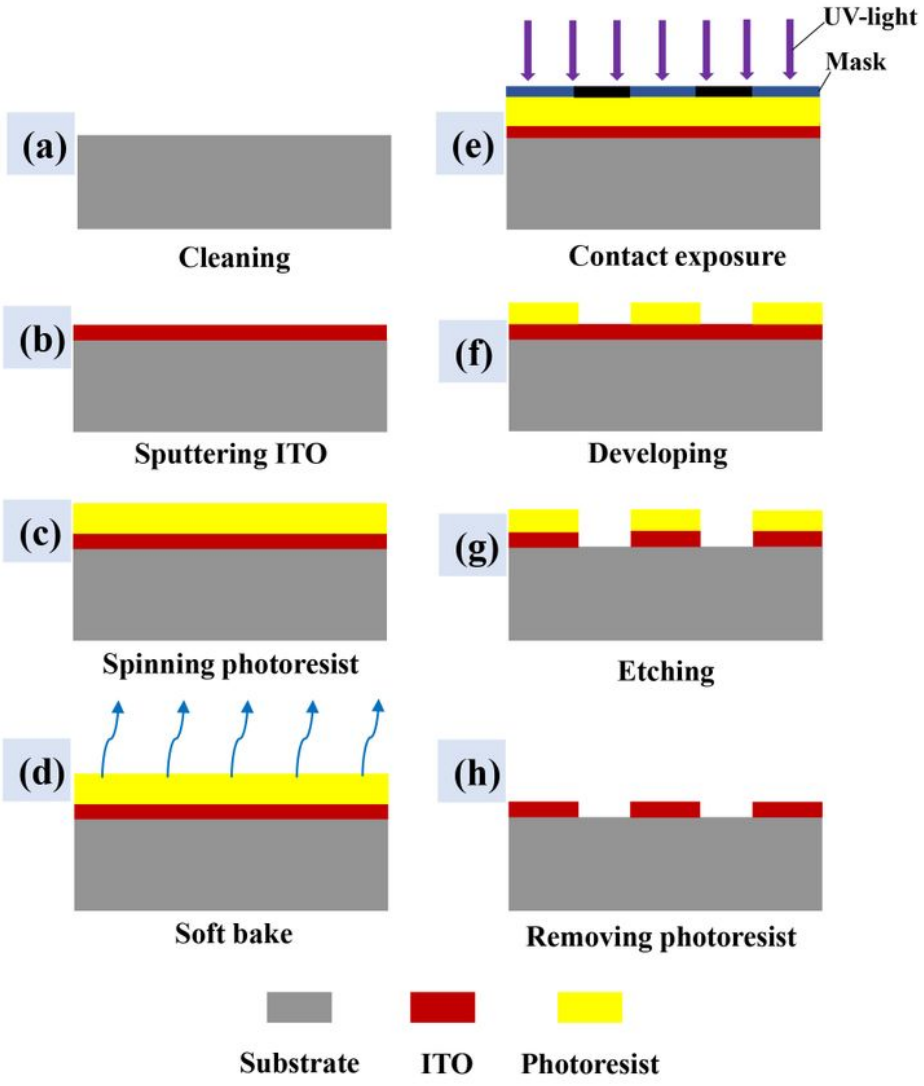


Figure 1

The micro fabrication process for the ITO TFSGs.

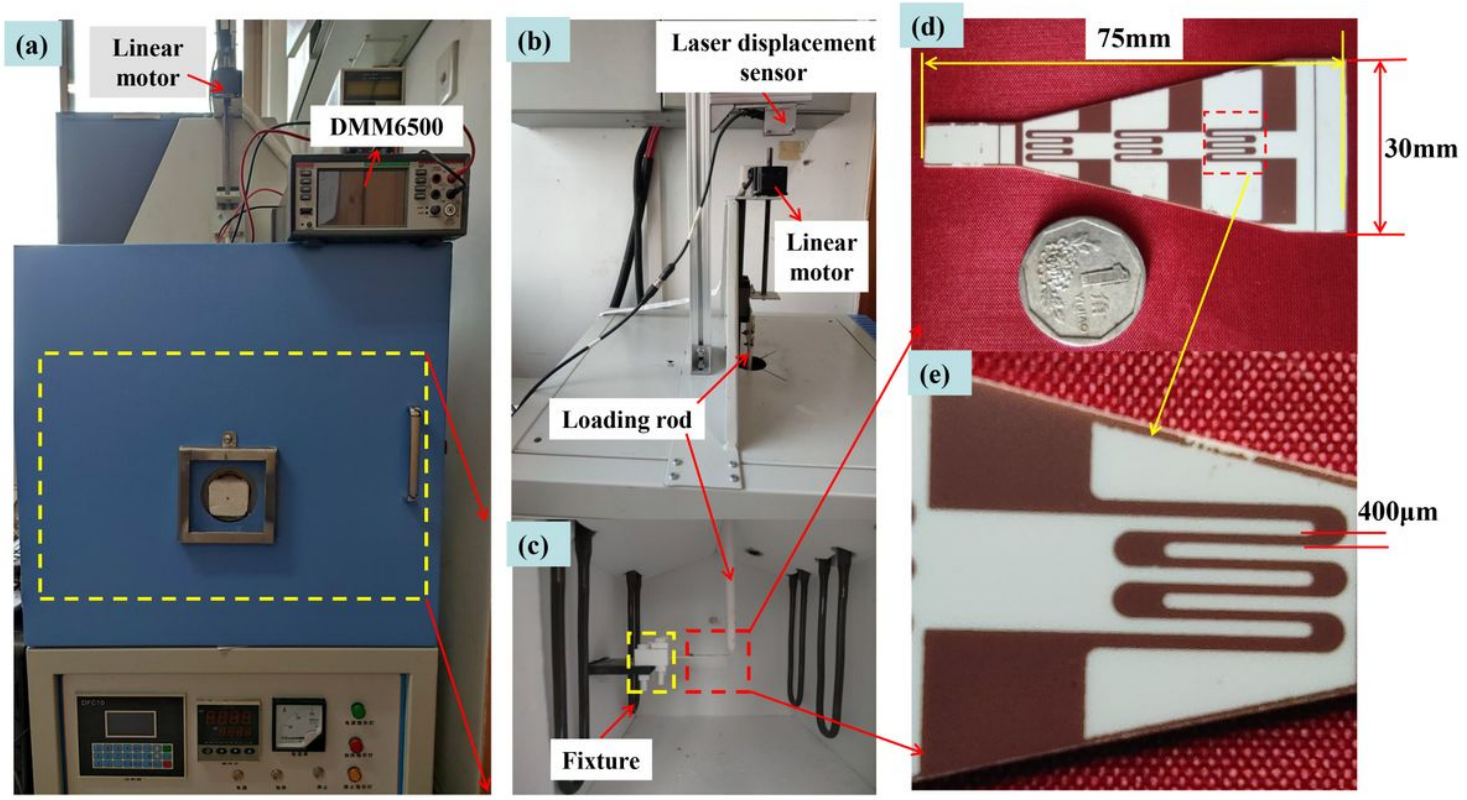


Figure 2

(a), (b), (c) The experimental setup for strain tests of the ITO TFSGs at elevated temperatures; (d) The morphology of the ITO TFSGs deposited on the alumina substrate. (e) The enlarged region of (d).

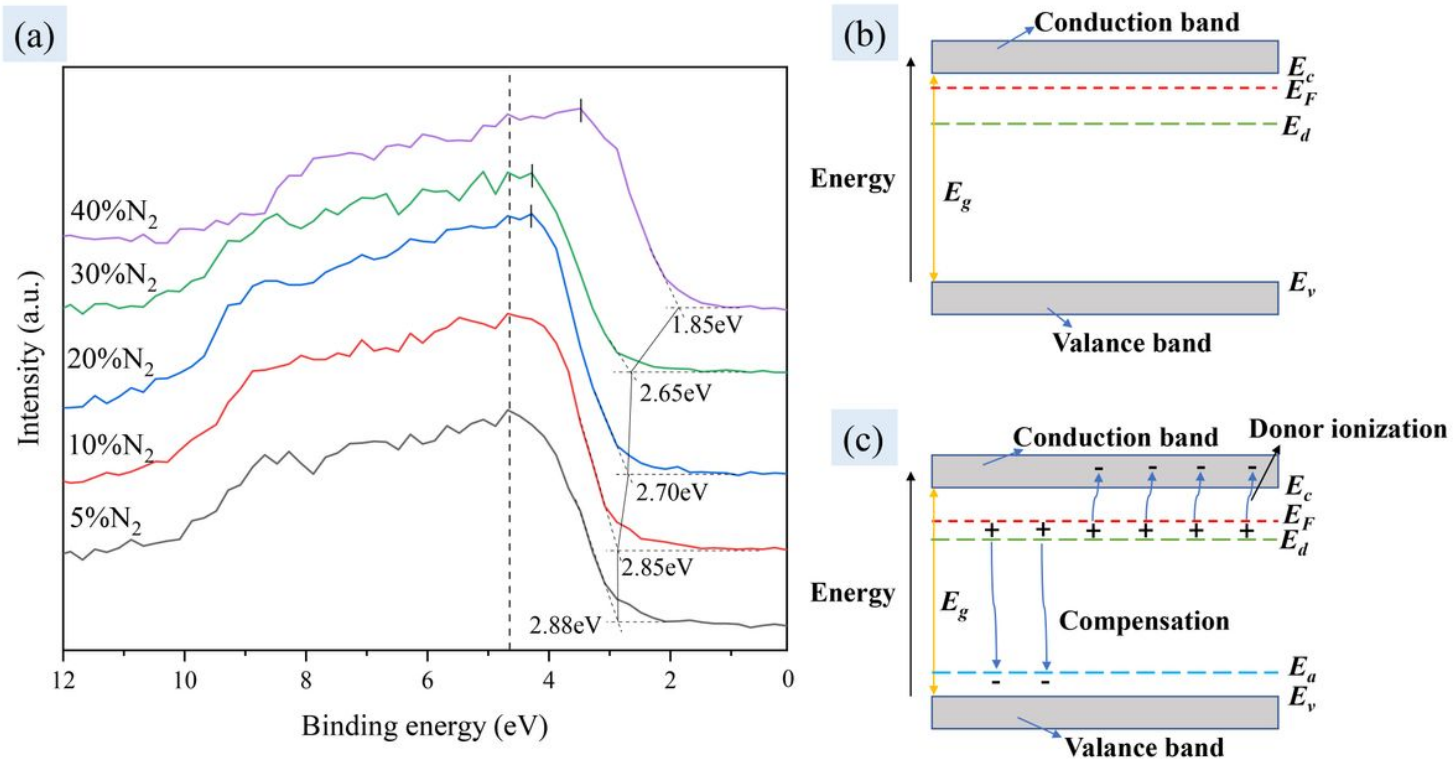


Figure 3

(a XPS valence band spectra of the ITO TFSGs prepared with different NPPs. b Diagrammatic sketch of energy level band of undoped ITO TFSGs. c Diagrammatic sketch of energy level band of nitrogen-doped ITO TFSGs.

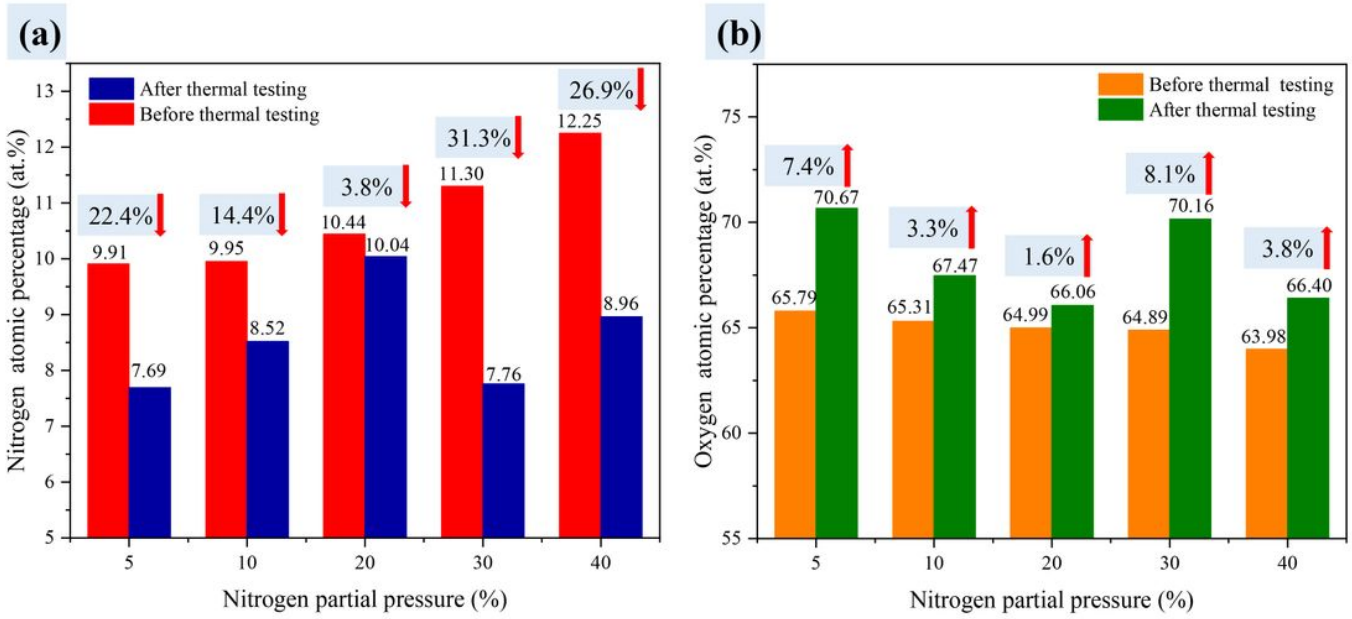


Figure 4

The elemental content analysis of the ITO TFSGs prepared with various NPPs: (a) the nitrogen atomic percentage before and after the thermal strain test; (b) the oxygen atomic percentage before and after the thermal strain test.

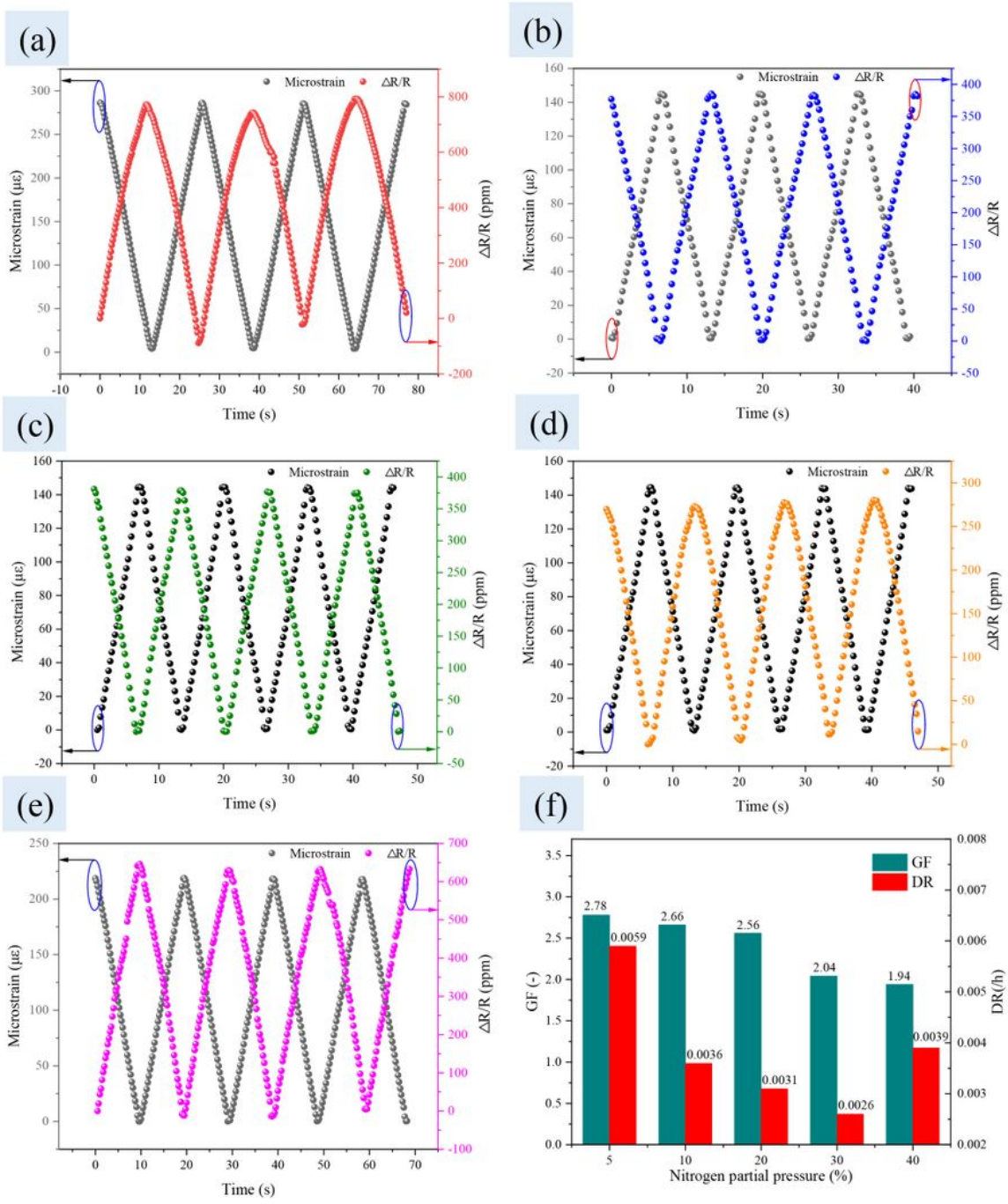


Figure 5

Piezoresistive response of (a) 5%N2 ITO, (b) 10%N2 ITO, (c) 20%N2 ITO, (d) 30%N2 ITO and (e) 40%N2 ITO TFSG during each loading-unloading cycle at room temperature, respectively. (f)The summaries of the GFs and DRs of the 5%~40N2 ITO TFSGs.

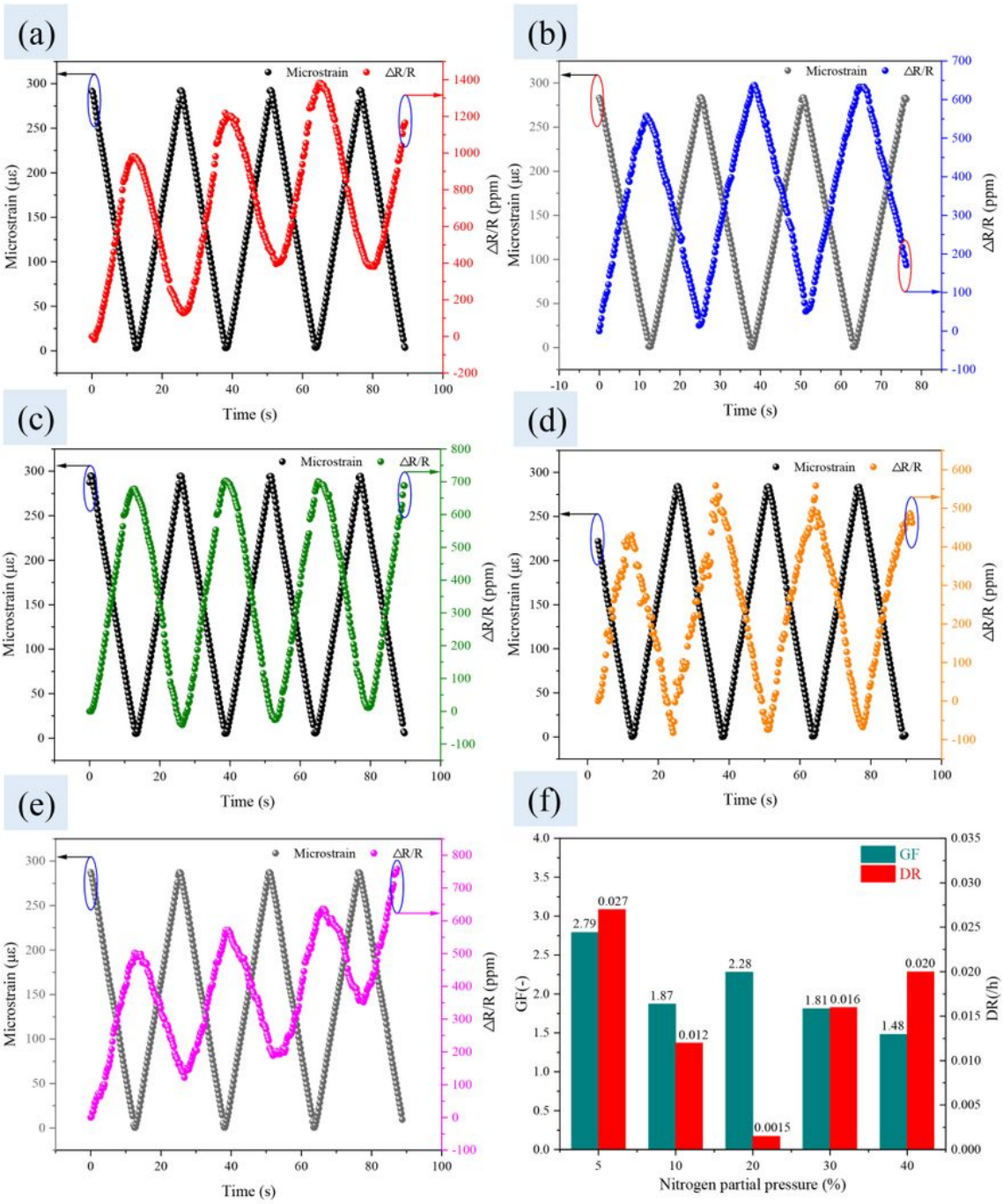


Figure 6

Piezoresistive response of (a) 5%N2 ITO, (b) 10%N2 ITO, (c) 20%N2 ITO, (d) 30%N2 ITO and (e) 40%N2 ITO TFSG during each loading-unloading cycle at 600℃, respectively. (f)The summaries of the GFs and DRs of the 5%~40N2 ITO TFSGs.

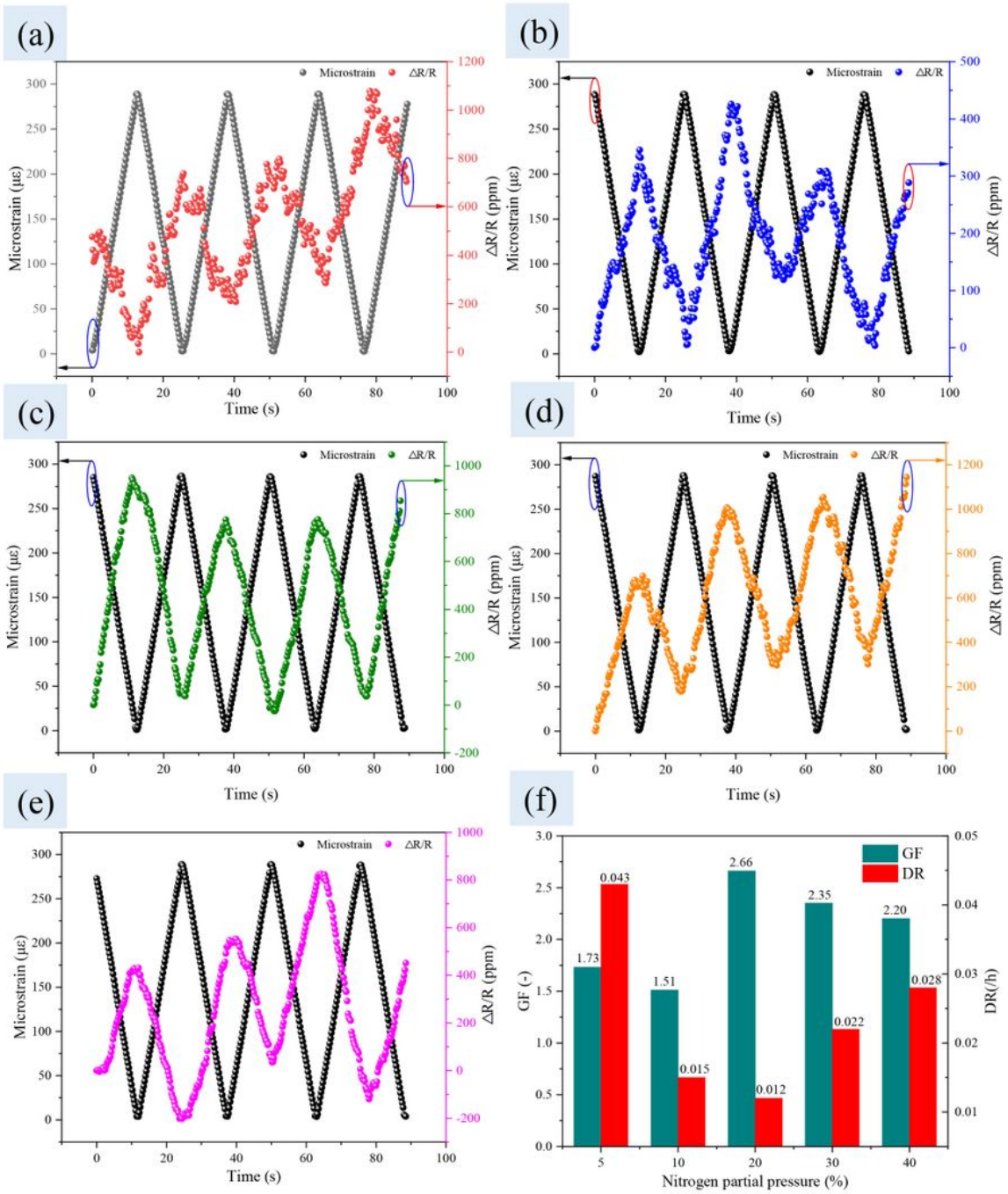


Figure 7

Piezoresistive response of (a) 5%N2 ITO, (b) 10%N2 ITO, (c) 20%N2 ITO, (d) 30%N2 ITO and (e) 40%N2 ITO TFSG during each loading-unloading cycle at 800℃, respectively. (f)The summaries of the GFs and DRs of the 5%~40N2 ITO TFSGs.

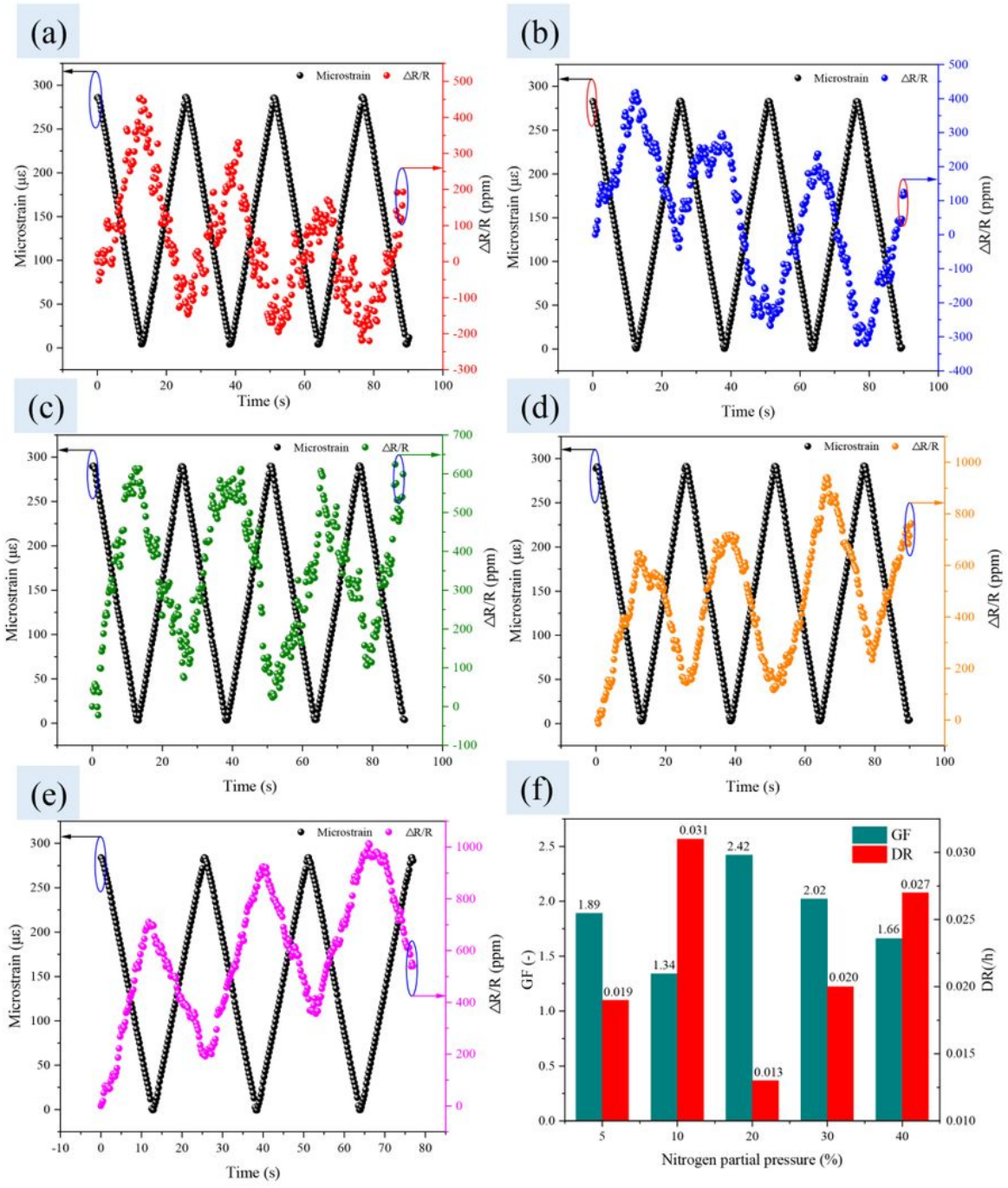


Figure 8

Piezoresistive response of (a) 5%N2 ITO, (b) 10%N2 ITO, (c) 20%N2 ITO, (d) 30%N2 ITO and (e) 40%N2 ITO TFSG during each loading-unloading cycle at 900℃, respectively. (f)The summaries of the GFs and DRs of the 5%~40N2 ITO TFSGs.

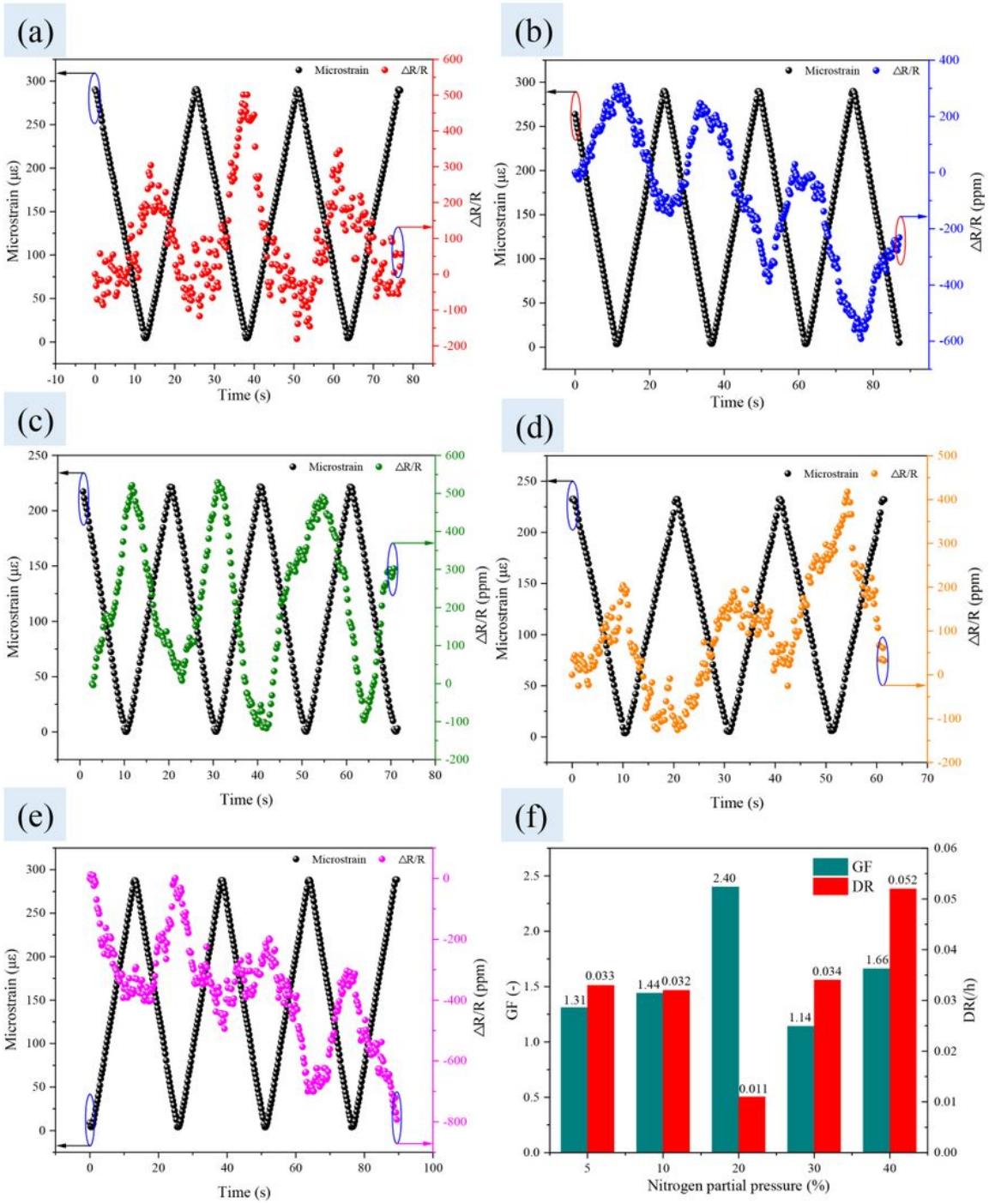


Figure 9

Piezoresistive response of (a) 5%N2 ITO, (b) 10%N2 ITO, (c) 20%N2 ITO, (d) 30%N2 ITO and (e) 40%N2 ITO TFSG during each loading-unloading cycle at 1000°C, respectively. (f)The summaries of the GFs and DRs of the 5%~40N2 ITO TFSGs.